

## Day 5 - Assessment: The Infrastructure Challenge

### Azure VM Networking Assessment - Step-by-Step Guide

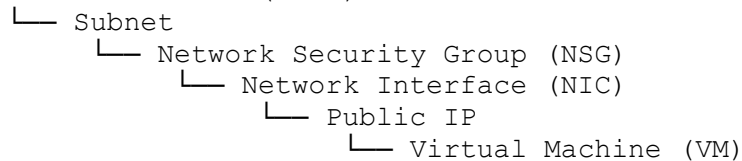
This document explains **end-to-end steps** to create an Azure Virtual Machine **from scratch**, using **separately created networking resources**. It also explains **why Azure creates default resources** and how to **replace them with your own**.

---

#### Architecture Overview

...

#### Virtual Network (VNet)



...

---

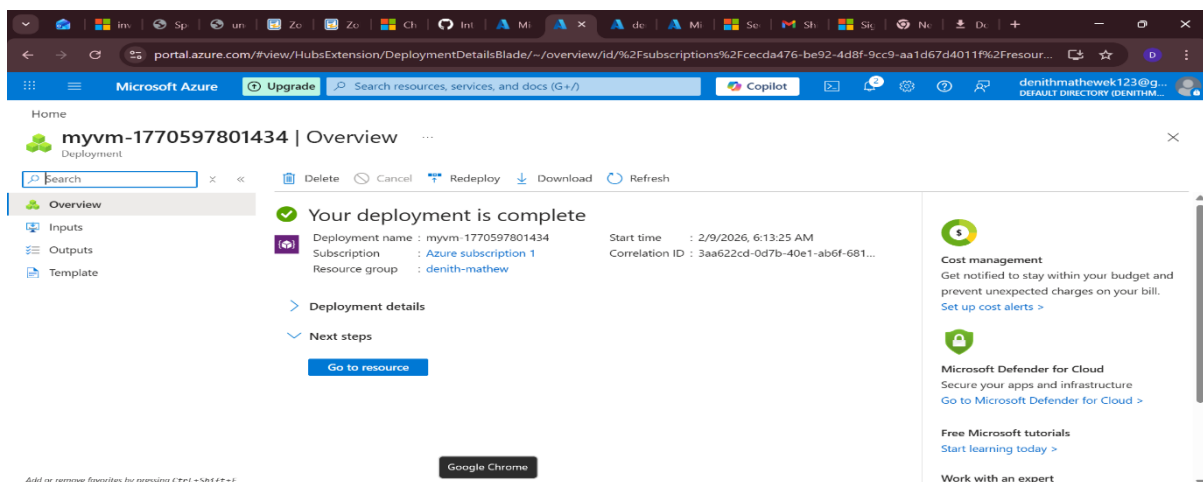
#### Task 1: Create Virtual Network (VNet) and Subnet

##### Purpose

- VNet provides a private network for Azure resources.
- Subnet logically divides the VNet and hosts the VM.

##### Steps

1. Go to **Virtual Networks** → **Create**
2. Select:
  - Resource Group
  - Region
3. VNet details:
  - Name: `vnet-lab`
  - Address space: `10.0.0.0/16`
4. Subnet configuration:
  - Subnet name: `subnet-vm`
  - Address range: `10.0.1.0/24`
5. Review and Create



## Task 2: Create Network Security Group (NSG)

### Purpose

- NSG acts as a **firewall** to control inbound and outbound traffic.

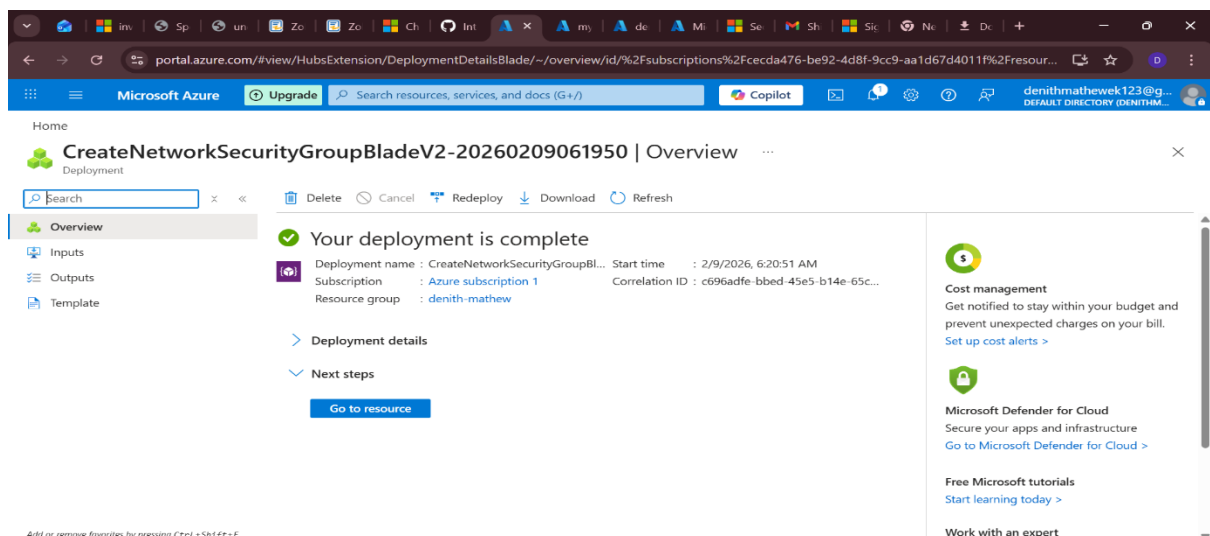
### Steps

1. Go to **Network Security Groups** → **Create**
2. Enter:
  - Name: `nsg-vm`
  - Resource Group
  - Region
3. Create the NSG

### Add Inbound Rules (Example)

- Allow SSH (Linux):
  - Port: 22
  - Protocol: TCP
  - Action: Allow
  - Priority: 100
- Allow RDP (Windows):
  - Port: 3389
  - Protocol: TCP
  - Action: Allow

Outbound rules are allowed by default unless restricted.



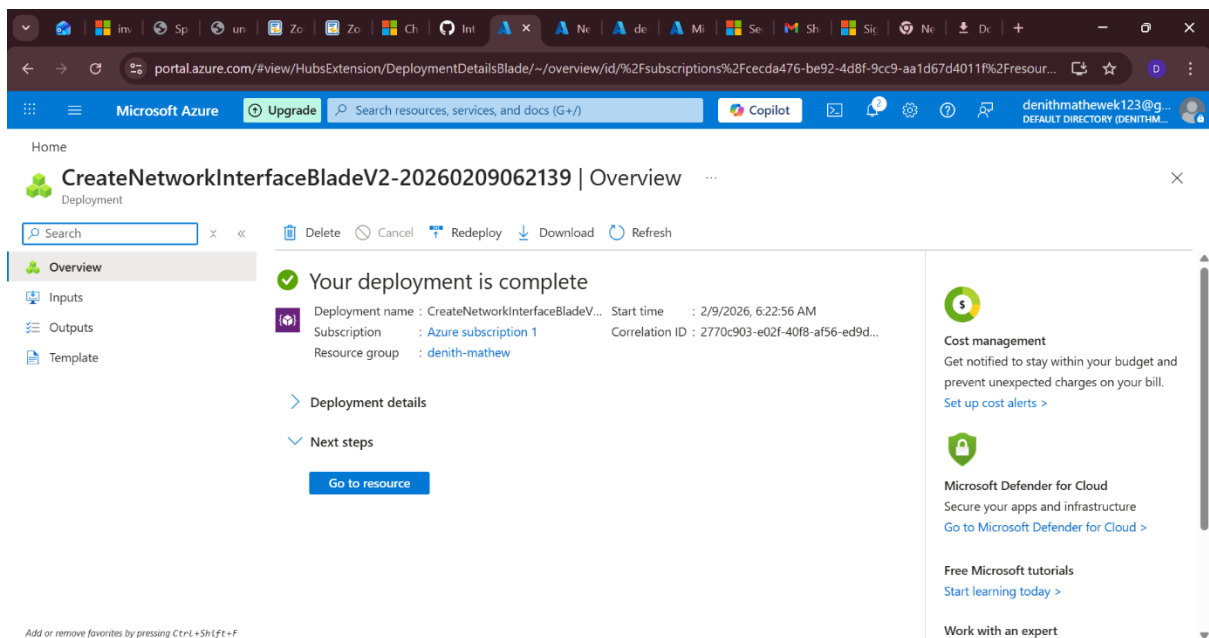
### Task 3: Create Network Interface (NIC)

#### Purpose

- NIC connects the VM to the subnet, NSG, and Public IP.

#### Steps

1. Go to **Network Interfaces** → **Create**
2. Enter:
  - Name: `nic-vm-custom`
  - Resource Group
  - Region
3. Networking:
  - Virtual Network: `vnet-lab`
  - Subnet: `subnet-vm`
  - Network Security Group: `nsg-vm`
4. Do NOT attach Public IP yet (optional)
5. Create NIC



## Task 4: Create Public IP Address

### Purpose

- Public IP allows external access to the VM.

### Steps

1. Go to **Public IP addresses** → **Create**
2. Enter:
  - Name: `pip-vm-custom`
  - IP version: IPv4
  - Assignment: Static (recommended)
3. Create Public IP

The screenshot shows the Azure portal interface for a deployment named 'PublicIPAddress-ARM'. The deployment is complete, as indicated by a green checkmark and the text 'Your deployment is complete'. The deployment details show it was created on 2/9/2026 at 6:24:21 AM, using 'Azure subscription 1' and the 'denith-mathew' resource group. The correlation ID is 0635a939-ecfc-4591-8b15-a0b... The left sidebar shows the 'Overview' tab selected, with links to 'Inputs', 'Outputs', and 'Template'. The right sidebar contains links for 'Cost management', 'Microsoft Defender for Cloud', and 'Free Microsoft tutorials'. A 'Go to resource' button is visible below the deployment details.

## 2.disk creation

The screenshot shows the Azure portal interface for a deployment named 'CreateDiskBlade-20260209064045'. The deployment is complete, as indicated by a green checkmark and the text 'Your deployment is complete'. The deployment details show it was created on 2/9/2026 at 6:41:35 AM, using 'Azure subscription 1' and the 'denith-mathew' resource group. The correlation ID is 88d4cae0-ccf5-446c-96c4-b433... The left sidebar shows the 'Overview' tab selected, with links to 'Inputs', 'Outputs', and 'Template'. The right sidebar contains links for 'Cost management', 'Microsoft Defender for Cloud', and 'Free Microsoft tutorials'. A 'Go to resource' button is visible below the deployment details.

## Task 5: Configure Resource Associations

### 1. Associate NSG with Subnet

1. Open `nsg-vm`
2. Go to **Subnets** → **Associate**
3. Select:
  - VNet: `vnet-lab`
  - Subnet: `subnet-vm`

### 2. Attach Public IP to NIC

1. Open `nic-vm-custom`
2. Go to **IP configurations** → `ipconfig1`
3. Associate Public IP: `pip-vm-custom`
4. Save

### Result

- NIC is now connected to Subnet, NSG, and Public IP

## IMPORTANT NOTE (Before Creating the VM)

### Why Azure Creates a NIC and Public IP by Default

#### When creating a VM through the Azure Portal:

- Azure automatically creates a **NIC and Public IP** to ensure:
  - Safe defaults
  - Simpler user experience
  - Guaranteed connectivity

The portal does **not** allow selecting an existing NIC during VM creation to avoid:

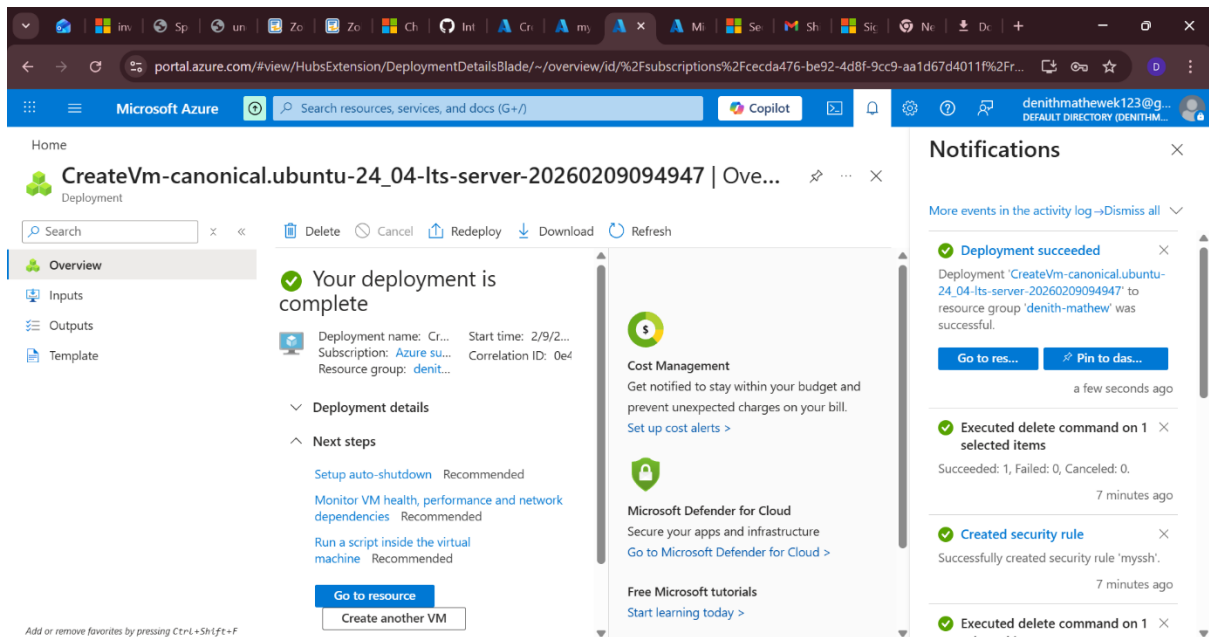
- Configuration conflicts
- Zone or subnet mismatch
- Lifecycle ownership issues

Advanced tools like **Azure CLI, PowerShell, and Terraform** allow attaching existing NICs because they operate closer to the Azure Resource Manager (ARM) API.

## Task 6: Create Virtual Machine

### Steps

1. Go to **Virtual Machines** → **Create**
2. Enter:
  - VM name: `vm-lab`
  - Image: Ubuntu / Windows
  - Size: as required
3. Networking tab:
  - Allow Azure to create default NIC and Public IP
4. Review and Create VM



## Task 7: Replace Default NIC with Custom NIC

### Purpose

- To ensure the VM uses **your pre-created networking resources**.

### Steps

1. Stop the VM
  - Open VM → **Stop** (Deallocate)
2. Detach Default NIC
  1. Go to VM → **Networking**
  2. Identify auto-created NIC
  3. Detach the NIC
3. Attach Custom NIC
  1. Still under VM Networking
  2. Attach NIC: ``nic-vm-custom``
  3. Save
4. Start the VM
  - Start VM after NIC replacement

## Task8 : Resource visualizer

Microsoft Azure portal interface showing the Resource visualizer for the resource group 'denith-mathew'.

The interface includes a sidebar with navigation options: Overview, Activity log, Access control (IAM), Tags, **Resource visualizer** (selected), Events, Settings, Cost Management, Monitoring, Automation, and Help.

The main area displays a resource diagram showing the following components and their relationships:

- Subscription (Root)
- Resource Group (denith-mathew)
- Storage Account (denith-mathew-storage)
- Virtual Machine (denith-mathew-vm)
- Virtual Machine Scale Set (denith-mathew-vmss)
- Virtual Machine Image (denith-mathew-vmimg)
- Virtual Machine Disk (denith-mathew-vmimg-disk)
- Virtual Machine Network Interface (denith-mathew-vmnic)
- Virtual Machine Network Interface Card (denith-mathew-vmnic-card)
- Virtual Machine Network Interface Adapter (denith-mathew-vmnic-adapter)

Buttons at the top of the main area include: Search, Choose resources, Reset diagram, Zoom to fit, Refresh, Export, and List view.

At the bottom, there is a 'Give feedback' section with a link to 'Tell us about your experience with resource visualizer'.

Footer text: Add or remove favorites by pressing Ctrl+Shift+F